

Predicting Dinitrogen Activation and Coupling with Carbon Dioxide and Other Small Molecules by Methyleneborane: A Combined DFT and Machine Learning Study

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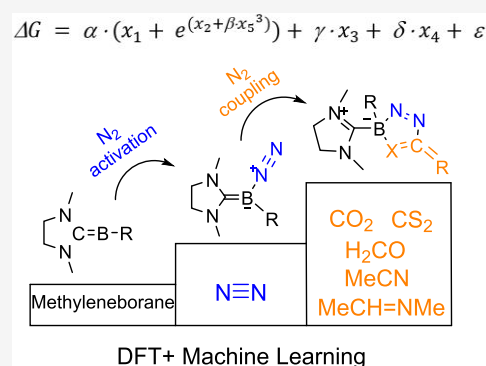
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ABSTRACT: The capture of carbon dioxide is extremely important due to the increasingly severe greenhouse effect, and the conversion of dinitrogen into high-value N–C compounds is of great significance. Here, we predict through density functional theory calculations that the coupling of dinitrogen with carbon dioxide by methyleneborane becomes favorable both thermodynamically and kinetically. Machine learning analysis suggests that increasing the HOMO–LUMO gap or the charge on the boron atom or decreasing the charge of the nitrogen atom will reduce the reaction energies. N₂ coupling is examined as a concerted step, as CO₂ is a σ donor and a π acceptor. Besides, N₂ coupling with other small molecules including formaldehyde, sulfur dioxide, acetonitrile, and N-ethylidenemethylamine was also examined in both thermodynamics and kinetics. Our findings highlight a crucial role of main group species in N₂ coupling chemistry.



INTRODUCTION

Carbon dioxide (CO₂) is an easily available and renewable C1 source. Nowadays, a sharp increase in anthropogenic CO₂ emissions has significantly exacerbated the greenhouse effect, leading to global warming.¹ Carbon capture and sequestration (CCS) technologies have become an efficient method to reduce global warming until the infrastructure is established with cleaner energies instead of carbon-based sources.^{2–4} A frustrated Lewis pair (FLP), containing a Lewis acid and a Lewis base in two separated sites, is a metal-free concept for novel catalysis and reactivity.^{5–12} It can be widely used to activate CO₂ owing to the electrophilicity of the carbon atom and the nucleophilicity of the oxygen atom.^{13–16} In addition, carbene,¹⁷ silylene,^{18–20} and germylene^{21,22} are other main group species to capture and activate CO₂.

Dinitrogen (N₂) is an abundant and easily accessible gas in the atmosphere. It is challenging to activate N₂ due to the significant HOMO–LUMO gap of 10.8 eV (HOMO: the highest occupied molecular orbital; LUMO: the lowest unoccupied molecular orbital) and N≡N bond strength with a bond dissociation energy of 945 kJ/mol.²³ The conversion of N₂ to high-value N–C compounds by a transition metal is common and important; the Holland group reported the reaction of N₂ coupling with benzene in 2020 (Figure 1a).²⁴ The early discovery of [N₂]^{4–} ligands in transition metal dinitrogen compounds through four-electron reduction supports the potential of N–C bond formation by N₂ coupling with CO₂.^{25–27} Though CO₂ also exhibits high stability in both thermodynamics and kinetics due to the high oxidation state of carbon,²⁸ the coupling of N₂ and CO₂ under

electro- or photocatalysis is broadly reported.^{29–37} The example of N₂ coupling with CO₂ by a transition metal can be traced back to 1976.³⁸ After that, lots of transition metals like zirconium (Zr),^{39,40} hafnium (Hf),⁴¹ titanium (Ti) (Figure 1b),^{42–46} niobium (Nb),^{47,48} molybdenum (Mo), and tungsten (W)^{49,50} are usually used to promote the formation of the C–N bond. However, the coupling of N₂ with CO₂ involving the main group compounds is limited. The Zhu group achieved N₂ coupling with CO₂ by a metal-free pincer complex in a theoretical study.⁵¹

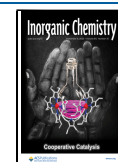
Our group predicted a dinitrogen reduction and explored the reaction mechanism theoretically by methyleneboranes.⁵² These methyleneborane–N₂ complexes have not yet been synthesized, but they resemble the dinitrogen-activated intermediates reported in *Science*.^{53,54} Can these thermodynamically favorable methyleneborane–N₂ complexes achieve N₂ coupling with CO₂ and other small molecules? Due to our ongoing interest in dinitrogen coupling and CO₂ activation, we carry out density functional theory (DFT) calculations to examine the feasibility of the capture of CO₂ and other small molecules by methyleneborane–N₂ complexes (Figure 1c). With the development of artificial intelligence technology,

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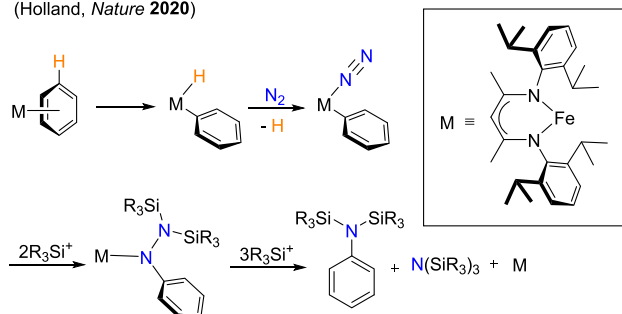
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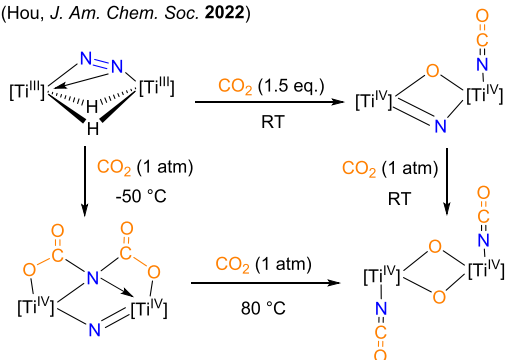
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(a) N₂ coupling with benzene by iron compounds
(Holland, *Nature* 2020)



(b) N₂ coupling with CO₂ by dititanium dihydride framework
(Hou, *J. Am. Chem. Soc.* 2022)



(c) N₂ coupling by methyleneborane-N₂ complexes in this work

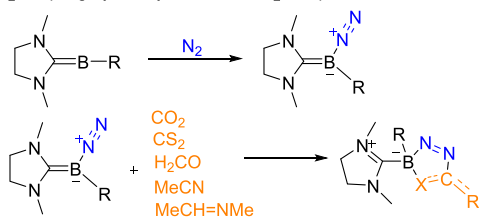


Figure 1. (a) Previous work in N₂ coupling with CO₂ by a dititanium dihydride framework. (b) Previous work in N₂ activation by methyleneboranes. (c) Probing N₂ coupling by methyleneborane–N₂ complexes.

machine learning has been applied to predicting molecular properties,^{55–59} chemical reactions,^{60–63} and materials science fields.^{64–67} Therefore, we also use it to examine the factors of the reaction energies.

COMPUTATIONAL DETAILS

All of the structures were optimized in the gas phase at the ω B97X-D level of density functional theory⁶⁸ using the standard cc-pVDZ basis set^{69,70} for all of the atoms. Frequency calculations at the same level of theory were performed to confirm that all stationary points were minima (no imaginary frequencies) or transition states (one imaginary frequency). Calculations of the intrinsic reaction coordinates (IRC)^{71,72} were also performed on the transition states to confirm that such structures indeed connected two minima. Single-point energy calculations were performed with the same functional level and a triple- ζ cc-pVTZ basis set.^{69,70} The relative Gibbs free energy (ΔG) is determined by adding the single-point energy to the Gibbs free energy correction acquired from the ω B97X-D/cc-pVDZ method. Fuzzy atom bond order (FBO) is calculated by the Multiwfn program.⁷³ PIO analysis is carried out by the available code on GitHub (<https://github.com/jxzhangcc/PIO>) combined with the NBO package⁷⁴ for natural

population analysis. The PIO method was developed by Lin and co-workers to better understand interactions between chemical fragments in a structure.⁷⁵ It can quantify the strength of each orbital interaction through the PIO-based bond index (PBI).⁷⁶ All of the optimizations were performed with the Gaussian 16 software package at 298 K without any constraints and using an ultrafine grid.⁷⁷

RESULTS AND DISCUSSION

Probing Methyleneborane–N₂ Complexes for N₂ Coupling with CO₂. In order to locate the optimal R substituents in boron, we selected the thermodynamically favorable methyleneborane–N₂ complexes **1-R** (R = CN, H, Me, Ph, PMe₂, SiMe₃, ^tBu; Figure 2) in previous study to

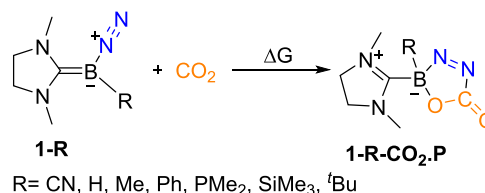


Figure 2. N₂ coupling with CO₂ by methyleneborane–N₂ complexes **1-R** containing different substituents.

examine the thermodynamic results of N₂ coupling with CO₂.⁵² Through geometric analyses, the distances between the boron atom and the nonadjacent nitrogen atom (N2) in N₂ of **1-R** are in the range of 2.553–2.562 Å, indicating rigid structures with a donor–acceptor interaction. As shown in Table 1, the C–O bond distances of the product are in the

Table 1. Bond Distances (Å) of **1-R-CO₂P** and Reaction Energies (ΔG , kcal/mol) of N₂ Coupling with CO₂

1-R	B–N	N–N	B–O	C–O	C–N2	ΔG
CN	1.564	1.238	1.460	1.341	1.494	2.0
H	1.552	1.242	1.473	1.335	1.491	0.3
Me	1.557	1.241	1.486	1.332	1.492	−6.5
Ph	1.554	1.240	1.482	1.333	1.493	−3.1
PMe ₂	1.540	1.245	1.475	1.338	1.485	−6.3
SiMe ₃	1.527	1.249	1.482	1.339	1.478	−1.3
^t Bu	1.553	1.242	1.489	1.331	1.491	−10.1

range of 1.331–1.341 Å, which are larger than that (1.163 Å) of the CO₂ molecule, and the reaction energies (ΔG) of N₂ coupling with CO₂ are in the range of −10.1–2.0 kcal/mol, indicating the potential of CO₂ activation.

The effect of *N*-heterocyclic carbenes (NHCs) for N₂ coupling with CO₂ was also considered. Based on previous work,⁵² we chose the thermodynamically favorable *n*-^tBu (*n* = 2, 3, 4, 5, 6) to couple with CO₂. Reaction energies are shown in Figure 3, and more information is shown in Table S1. Generally, the cyclic alkyl amino carbenes (CAACs) display less favorable N₂ coupling than cyclic diamino carbenes (CDACs). The CAACs contain a nitrogen and a four-coordinated carbon atom, acting as both a σ donor and a π acceptor. However, CO₂ is also a σ donor to the boron atom and a π acceptor to the N2 atom, indicating a competition with CAACs.

Principal interaction orbital (PIO) analysis developed by the Lin group, which can identify the bonding interaction in chemical imaging and quantify with the PBI, is carried out to

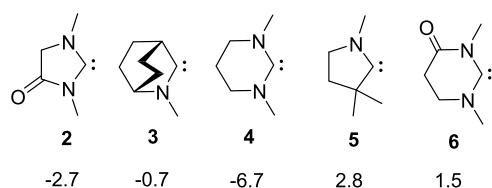


Figure 3. Selected NHCs of methyleneborane–N₂ complexes and reaction energies (ΔG , kcal/mol) of N₂ coupling with CO₂ by *n*-^tBu containing different NHCs are shown at the bottom.

see the bond interaction between the CO₂ moiety and the remaining moiety of the coupling product **1**-^tBu-CO₂-P.^{75,76} As shown in **Figure 4**, the first PIO pair indicates that the p

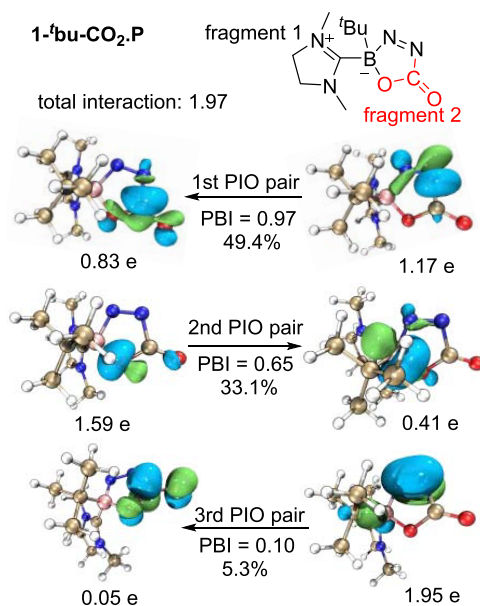


Figure 4. PIO analysis with the CO₂ moiety and the remaining moiety as two fragments. The top two PIOs of **1**-^tBu-CO₂-P. PBI and its contribution (as %) to the total interactions between two fragments (the contributions of all PIOs sum up to 100%). The isovalue is 0.05 au.

orbital of the N₂ atom donates electrons to the π^* orbital of the C–O bond of CO₂ with a PBI of 0.97. The second PIO pair is σ -donation from the lone-pair electrons of the oxygen atom to the vacant p orbital of the boron atom, and the PBI is 0.65. The third PIO pair is the π -backdonation from the π orbital of dinitrogen to the π^* orbital of the C–O bond with a PBI of 0.10.

Preliminary processing of the data was performed using the MATLAB (Version: 24.2.0.2923080 (R2024b) Update 6) software package to probe the factors on the reaction energies. Through the analysis of the linear correlation among ΔG , HOMO–LUMO gap, natural population analysis (NPA) charge of the boron and nitrogen atom (NPA_B, NPA_{N1}, NPA_{N2}), and the fuzzy atom bond order difference (ΔFBO) of all bonds between the products and reactants, (represented as y and x_1 to x_5 , respectively; **Figure 5**), it is observed that variables 3 and 4 have an extremely high linear correlation with each other, whereas others show very low correlation with y . We propose an expression to model the relationship, which consists of both linear components $L(x)$ and nonlinear components $N(x)$, formulated as follows.

$$\Delta G = L(x_3, x_4) + N(x_1, x_2, x_5) \quad (1)$$

To explore the nonlinear relationships present in the data, symbolic regression as one of the supervised machine learning methods searches the space of mathematical expressions to find the model that best fits a given data set, in terms of both accuracy and simplicity. Unlike methods reliant on prior assumptions, symbolic regression operates without such presuppositions, autonomously discovering fundamental mathematical expressions directly from data. This approach enables the discovery of complex, nonlinear patterns directly from the data. PySR⁷⁸ is an efficient and open-source software package specifically developed for conducting symbolic regression. The symbolic regression of the above data using Python and the PySR package, according to **eq 1**, yields the following specific formula

$$\Delta G = \alpha \cdot (x_1 + e^{(x_2 + \beta \cdot x_5^3)}) + \gamma \cdot x_3 + \delta \cdot x_4 + \varepsilon \quad (2)$$

Root mean square error (RMSE) is a measure used to evaluate the accuracy of a model by quantifying the differences between predicted and observed values, calculated as the square root of the average of squared errors. RMSE of this model is 1.9, which is lower than the 2.2 obtained from the linear regression model (see **Table S2**). This suggests that there is indeed a nonlinear relationship between them. According to the coefficients (**Table 2**), it is evident that increases in x_1 , x_2 , and x_5 lead to a decrease in y . In other words, increasing the HOMO–LUMO gap, NPA_B, and ΔFBO reduces ΔG . Conversely, increasing NPA_{N1} and NPA_{N2} results in an increase in ΔG . Although this reaction is an entropy-decreasing reaction, stronger bonds are formed, making it a thermodynamically exothermic reaction. In addition, in comparison with three other commonly used machine learning models, symbolic regression still maintains the highest accuracy (**Figure S1**).

Mechanistic Studies. All thermodynamically favorable **1**-**R** (**R** = Me, Ph, PMe₂, SiMe₃, ^tBu) are selected for further mechanistic studies (**Figure 6**). The N₂ coupling involves a concerted step. In contrast, N₂ activation starts with a Lewis acid and then a Lewis base.^{79,80} The reaction barrier is consistent with the reaction energy. To probe the origin of the reaction barrier, energy decomposition analysis (EDA), known as the distortion/interaction model and activation strain model,^{81,82} is carried out on the transition states (**Table 3**). The deformation energies of reactants and binding of deformed reactants (ΔE_b) destabilize and stabilize the transition states, respectively, as shown in the equation $\Delta E(\text{barrier}) = \Delta E_{\text{deform}}(\text{reactants}) + \Delta E_b$. The deformation energies of **1**-**R** are in the range of 21.0–29.8 kcal/mol, while those of CO₂ range from 33.4 to 39.4 kcal/mol. The deformation energy of the reactants of **1**-SiMe₃-CO₂-TS is the highest, leading to the highest reaction barrier. The lowest reaction barrier of **1**-^tBu-CO₂-TS could be attributed to the largest binding of deformed reactants (−47.7 kcal/mol).

N₂ Coupling with Other Small Molecules. Besides CO₂, other small molecules, such as sulfur dioxide, formaldehyde, acetonitrile, and *N*-ethylidenemethylamine, are also investigated for the coupling reactions. Through PIO analyses in transition states, it is the combination of the occupied p orbital in the nitrogen atom and the empty orbital in the boron atom that contributes to the C–X (O, S, N) polar bond activation (**Figure S2**). As shown in **Figure 7**, the reaction barrier for activating small molecules is in the order of formaldehyde

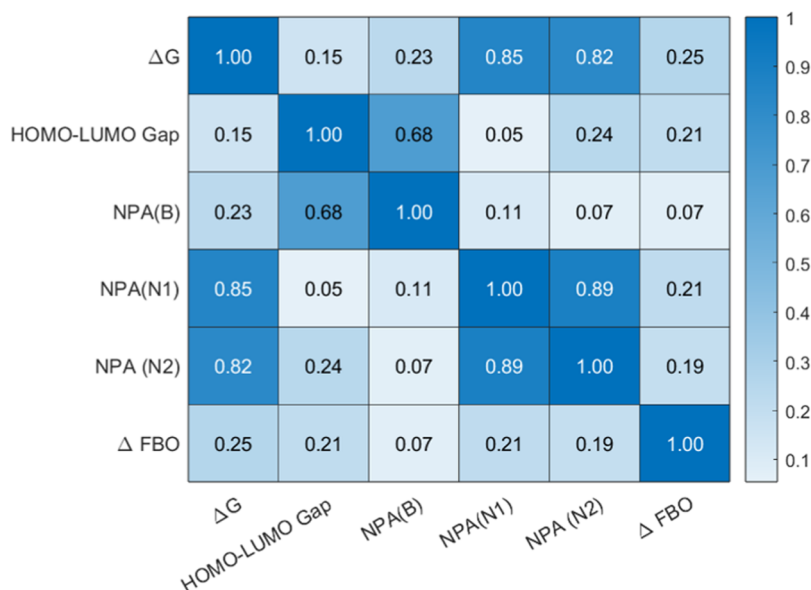


Figure 5. Heatmap of the absolute value of the Pearson correlation.

Table 2. Coefficients for the Nonlinear Model

variable	α	β	γ
estimate	−13.8091	6.2150	158.1546
variable	δ	ε	
estimate	51.7208	120.2445	

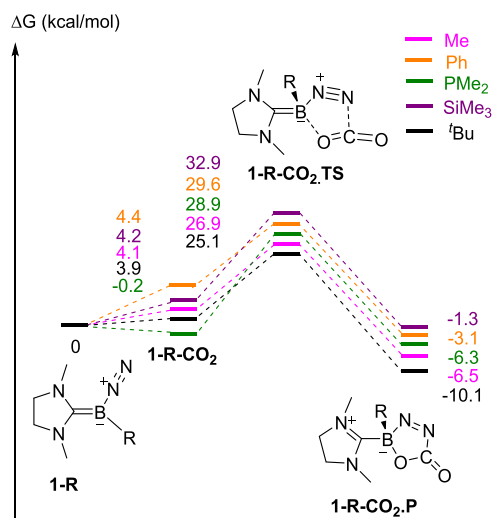


Figure 6. Gibbs energy profile for the CO₂ activation by 1-R (kcal/mol).

Table 3. Energy Decomposition Analysis on 1-R-CO₂.TS^a

1-R-CO ₂ .TS	ΔE_{deform} (1-R)	ΔE_{deform} (CO ₂)	ΔE_b	ΔE
Me	21.4	33.4	−38.5	16.2
Ph	22.3	33.9	−37.6	18.6
PMe ₂	21.0	35.1	−38.5	17.6
SiMe ₃	29.8	39.4	−47.5	21.8
tBu	24.0	36.8	−47.7	13.1

^aThe energies are given in kcal/mol.

(11.9 kcal/mol)<sulfur dioxide (19.2 kcal/mol)<carbon dioxide (25.1 kcal/mol)<acetonitrile (25.7 kcal/mol)<*N*-

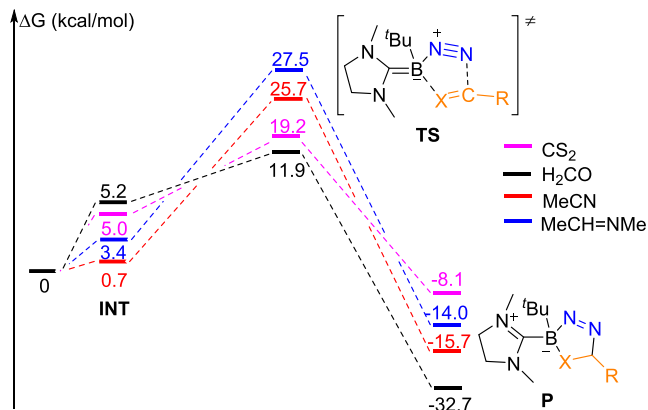


Figure 7. Gibbs energy profile for the activation of small molecules (kcal/mol).

ethylidenemethylamine (27.5 kcal/mol). EDA is carried out to probe the origin of the activation energy, as shown in Table 4. The difference in formaldehyde and *N*-ethylidenemethyl-

Table 4. Energy Decomposition Analysis on Transition States for Small Molecule Activation^a

	ΔE_{deform} (1-Bu)	ΔE_{deform} (small molecules)	ΔE_b	ΔE
CO ₂	24.0	36.8	−47.7	13.1
CS ₂	9.4	15.5	−16.8	8.1
H ₂ CO	13.6	4.3	−19.6	−1.8
MeCN	19.7	9.4	−15.7	13.3
MeCH=NMe	22.0	9.0	−18.8	12.1

^aThe energies are given in kcal/mol.

amine activation, with the lowest and highest activation energies, respectively, lies in reactant deformation energy rather than binding energy.

CONCLUSIONS

In summary, we investigated the N₂ coupling with CO₂ by methyleneborane–N₂ complexes 1-R and explored the effects

of the substituent and NHCs by DFT calculations and machine learning. Machine learning analysis suggests that increasing the HOMO–LUMO gap or the charge on the boron atom or decreasing the charge of the nitrogen atom will reduce the reaction energies. CDACs display more favorable N₂ coupling than CAACs because CAACs are both σ donors and π acceptors, forming a competition with CO₂. According to the kinetic study, the N₂ coupling is a concerted step, and the nitrogen atom donates electrons to the π^* orbital of the carbon atom of CO₂, with σ -donation from the lone-pair electrons of the oxygen atom to the vacant p orbital of the boron atom. **1**-⁴Bu performed best because of the largest binding of deformed reactants, which could be attributed to its highest HOMO energy of the fragment. Besides, N₂ coupling with other small molecules like sulfur dioxide, formaldehyde, acetonitrile, and *N*-ethylidenemethylamine was also examined in terms of both thermodynamics and kinetics. Although the substrate we designed has only been reported as an intermediate and its synthesis remains challenging, our findings provide insights into the N₂ coupling with small molecules by main group species. Hopefully, this proof-of-concept study will facilitate synthetic chemists' exploration of this challenging field.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.5c02384>.

Principal interaction orbital (PIO) analysis; energy decomposition analysis (EDA), and detailed information (PDF)

Cartesian coordinates (XYZ)

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Notes

The authors declare no competing financial interest.

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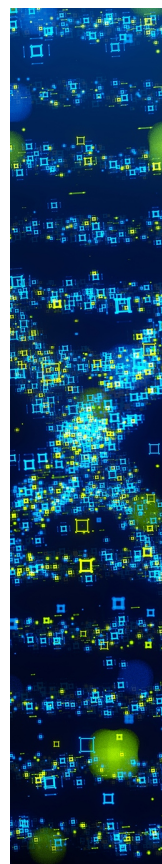
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